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Effective medium description of plasmonic couplings in disordered polymer and gold nanoparticle composites

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ABSTRACT

Disordered gold nanocomposite films were obtained by introducing spherical gold nanoparticles into a polymer matrix. These films were structurally characterized using microscopy techniques, and then analyzed for their effective optical indices using spectroscopic ellipsometry. It is found that even for a low volume fraction of gold nanoparticles ($f \sim 1-5\%$), the observed plasmonic resonance is affected by electromagnetic coupling between particles, related to the disorder of the particles. We show that the classical Maxwell Garnett Effective Medium Approximation (EMA) fails to predict the measured indices. Since couplings create deformations of the polarizability tensor of individual particles, we propose to take them into account phenomenologically using a modified Maxwell Garnett EMA based on a distribution of ellipsoidal polarizabilities. This modified model, albeit simple, appears appropriate: in simple cases, a unimodal distribution of ellipsoidal polarizabilities is used, allowing for good fits of the experimental data with only two free parameters. Bimodal distributions make it possible to handle more complex cases where the resonance presents a shoulder, suggesting that particles can then be categorized into weakly vs. strongly coupled resonators. Such modified Maxwell Garnett EMAs present the advantage of relying on physically meaningful parameters and could be a general tool for the phenomenological description of plasmonic couplings in different disordered nanocomposites.
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1. Introduction

The interest for nanostructures has exploded in the recent years, with many applications in view [1], from bio-medical sensing [2] to optical circuitry [3]. Metallic nanostructures, in particular, can display exciting properties, deriving from the presence of localized surface plasmons [1,4]. One growing class of plasmonic nanostructures is optical metamaterials, which have in common non-natural responses to electromagnetic waves [5,6,7,8,9]. Generally speaking, plasmonic nanostructures have mostly been produced using top-down methods based on nano-lithography [10,11]. New schemes are now emerging based on chemical synthesis and self-assembly of nanoparticles, allowing to envision 3-D, large area materials and relatively low-tech fabrication methods [12,13,14]. These assembly routes rely on colloidal interactions [15,16], finely dependent on the chemistry of the nanoparticles. Hence, often more than their lithographically produced counterparts, the self-assembled nanoparticle-based materials can present significant amounts of inhomogeneity in their organization, and therefore possible couplings between resonators even at low fraction. Extraordinary or meta-properties will only be attainable with nanoparticle assemblies if either the elementary nanoparticles present the right resonance [17,18], or their assembly produces the right coupled modes [19]. In any case, the assembly must be of high density, producing a material with a high metal volume fraction, which can be achieved by diverse routes. For such dense nanocomposites including some degree of disorder, measuring and modeling the optical index is not straightforward. Since long, spectroscopic ellipsometry has been used as a robust and extremely sensitive tool [20] for material characterization. During the last decade, its domain of application has been extended to the study of nanostructured samples [21,22,23,24,25]. The extraction of optical properties necessitates either the use of purely empirical generic dispersion functions [26,27,28], or alternatively, effective medium approximations (EMA) endowed with more physical content, such as the Maxwell Garnett [29] and Bruggeman [30] mixing laws. These are simpler to implement than full electrodynamics simulations, but their use is restricted to samples in which the size of the metallic inclusions is

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much smaller than the incident wavelength (quasistatic approximation). The Maxwell Garnett EMA (MG-EMA), for instance, accounts well for the local surface plasmon resonance of noble-metal nanoparticles (NPs) dispersed in a dielectric matrix, without interparticle couplings [26,27,31,32,33]. For higher nanoparticle concentrations, however, this approximation breaks down and modifications have to be introduced [34]. There is a large body of literature on how to extend the applicability of these effective medium theories to more complex samples [35,36,37,38,39,40,41,42,43]. These generalized effective medium theories are complex and their application to spectroscopic ellipsometry quickly becomes unpractical. Hence, other approaches are required that are mathematically simple but nevertheless give insights into the structure of the sample.

Here, we propose a phenomenological effective medium model to describe the optical properties of nanocomposites made of plasmonic nanoparticles inside a dielectric matrix, with some amount of electromagnetic coupling between nanoparticles. To test its validity, we produced samples consisting of a polymer film doped with spherical gold NPs with tunable NP volume fraction and film thickness. These are easier to manufacture than for instance Langmuir–Blodgett films [13] or microfluidic-controlled drying [14], but present similar features: plasmonic absorption, structural disorder and tunable optical and geometrical properties. Finally, we compared the optical properties obtained by spectroscopic ellipsometry with the prediction of our phenomenological model. We show that this model is simple to implement and agrees better with the measurements than the basic MG-EMA, by accounting for the electromagnetic couplings between NPs.

The manuscript is organized as follows. In the first part, we present the experimental methods and the description of the samples. In the second part, we present the optical measurements on the gold nanoparticle–polymer composites and show that real samples can have low-volume fractions (less than 6%) of dispersed particles, but nevertheless show optical properties significantly different from the ones given by the MG-EMA. In the third part, we show that these deviations are mainly due to coupling between close enough particles, and we propose a modification of the MG-EMA in order to account for these couplings in a phenomenological way.

2. Experimental system

2.1. Gold nanoparticles

Citrate-coated spherical gold nanoparticles were synthesized in water following the Turkevitch method [44,45,46], yielding a nanoparticle suspension of gold volume fraction $f = 2.5 \times 10^{-6}$. Their narrow size distribution was quantitatively analyzed by two methods. On the one hand, the nanoparticles were imaged by transmission electron microscopy (TEM) (Fig. 1a) and a statistical image analysis of the micrographs was performed. Here, the micrograph was obtained from a drop of colloid suspension dried onto a microscope grid. It can be seen that all nanoparticles are spherical and that some clustering occurs upon the drying process, without any merging between the nanospheres.

On the other hand, small-angle X-ray scattering (SAXS) data were obtained on the synthesis suspension. SAXS profiles were acquired in transmission on a Nanostar-U instrument (Bruker AXS) with a copper anode source, Goebel mirrors and a HiSTAR 2-D detector giving access to scattering vector q values from 0.1 to 8 nm^{-1}. The resulting 2-D image was found to be isotropic, and the data were azimuthally averaged to yield the scattering curve of the intensity corrected from experimental background versus $q = (4\pi/\lambda)\sin\theta$, where $\lambda = 0.154$ nm is the wavelength of the Cu Kα radiation and θ is half the scattering angle. The SAXS intensity $I(q)$ was fitted to the form factor [47] of a sphere of radius R (see Fig. 2):

$$P(q,R) = \frac{(\sin(qR) - qR\cos(qR))^2}{q^6} \tag{1}$$

convoluted with a Gaussian size distribution:

$$D(R,R_0) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(R-R_0)^2}{2\sigma^2}\right) \tag{2}$$

The Gaussian function found by SAXS presents a mean radius $R_0 = 6.5$ nm for the nanoparticles (with a standard deviation of 0.9 nm), and superposes well with the TEM-issued histogram of sizes (see Fig. 2).

Spectrophotometry, on the wavelength range 300–900 nm, was performed on the aqueous nanoparticle suspension issued from the synthesis, which presents no interparticle interactions. The nanoparticle absorption cross-section extracted from these data presents an intense and narrow plasmon absorption band at $\lambda = 520$ nm. This cross-section can be expressed as [48]:

$$\sigma = \frac{8\pi^2\sqrt{\varepsilon_m}}{\lambda} R^3 \mathrm{Im}\left(\frac{\varepsilon_{Au} - \varepsilon_m}{\varepsilon_{Au} + 2\varepsilon_m}\right) \tag{3}$$

where $\varepsilon_m = 1.769$ is the dielectric permittivity of water. When using, for the dielectric permittivity of gold ε_{Au}, the experimental Johnson and Christy dataset [49] ε_{JC}, we obtained the plot in dashed line in the Fig. 3. Compared to this plot, the experimental data show a slight broadening of the absorption band. We have checked that this broadening cannot be related, here, either to nanoparticle clustering, which does not occur in the dilute dispersion in water, or to size and shape polydispersities [50], which have a negligible effect for the studied nanoparticles. The broadening can therefore be attributed to a modification of the dielectric permittivity compared to that of bulk gold due to electron confinement in small particles.

To take this effect into account, we start by classically writing the bulk gold permittivity as the sum of two contributions [51], one

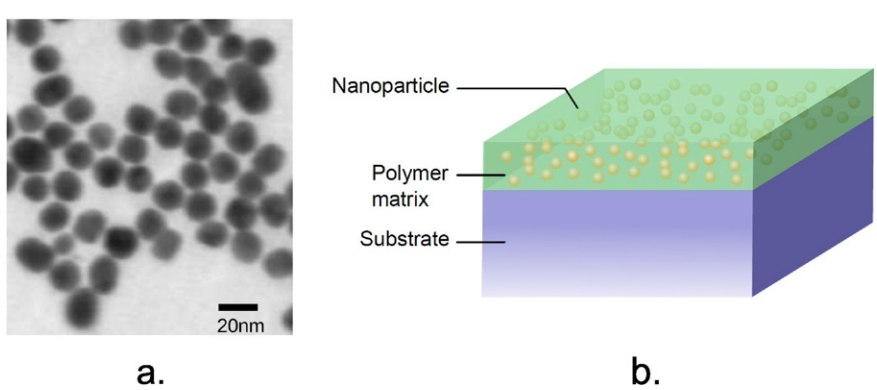

Fig. 1. Nanoparticles and composite films. a. Transmission electronic micrograph (TEM) of the gold nanoparticles b. Sketch of a nanocomposite thin film on a glass or silicon substrate after the spincoating process. No spatial order of the gold nanoparticles is expected in the thin film.

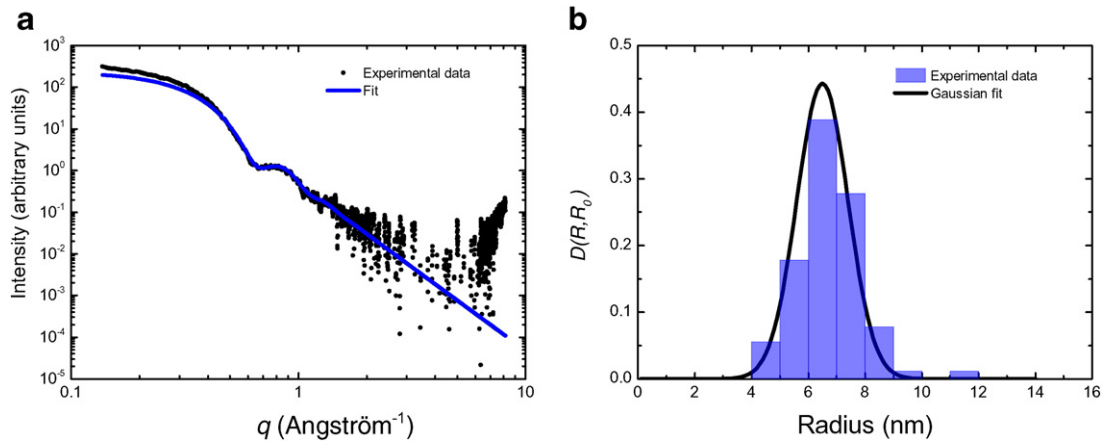

Fig. 2. (a) SAXS experimental data (dots) and adjusted form factor for an assembly of polydisperse spheres with a Gaussian size distribution (continuous line); (b) size distribution of the gold nanoparticles: Gaussian distribution extracted from the fit of the SAXS data in (a) (continuous line) and histogram extracted from the TEM micrograph analysis (blue bars).

stemming from interband transitions of bound electrons, denoted as ε_{IB}, and the other stemming from intraband transitions of free electrons that are well-described by the Drude model [52]:

$$\varepsilon_{Au}(\omega) = \varepsilon_{IB}(\omega) + \varepsilon_{Drude}(\omega). \quad (4)$$

The Drude contribution

$$\varepsilon_{Drude}(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma}, \quad (5)$$

is fitted to the Johnson and Christy dataset in the infrared ($\hbar\omega < 1.9$ eV), where the Drude contribution dominates, yielding the values for ω_p, the plasma frequency ($\hbar\omega_p = 8.92$ eV), and $\gamma = \gamma_0$, the bulk damping rate of gold ($\hbar\gamma_0 = 0.072$ eV). The interband contribution is then found from the same dataset, as the excess permittivity not accounted for by the Drude term:

$$\varepsilon_{IB}(\omega) = \varepsilon_{JC}(\omega) - \varepsilon_{Drude}. \quad (6)$$

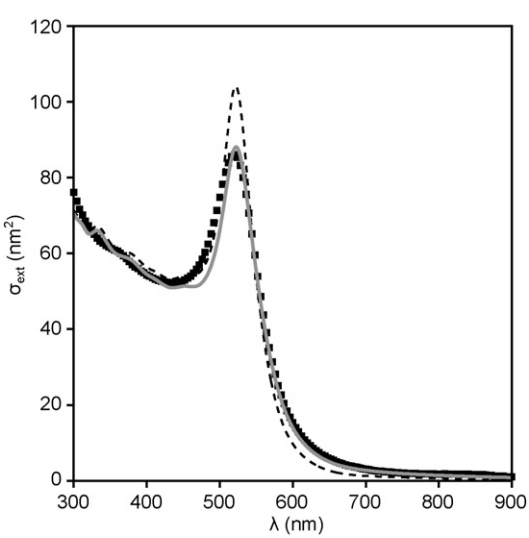

Fig. 3. Nanoparticle absorption cross-section. Squares: Experimental absorption cross-section of the gold nanoparticles in aqueous suspension. Lines: calculated absorption cross-section of the gold nanoparticles, using Eqs. (3) and (4), with the bulk gold permittivity (dashed line) or the modified permittivity taking electron confinement (finite-size) into account (continuous line).

For particles of small enough diameter, scattering of the electrons at the particle surface becomes important and reduces the mean free path; this can be taken into account by increasing the electron damping rate in the Drude contribution (Eq. (5)) by a surface-related amount γ_s, which depends on the size of the particle [53,51]: $\gamma = \gamma_0 + \gamma_s$. The surface damping rate γ_s can be determined experimentally, for the studied particles, by substituting Eq. (4) into Eq. (3), and then fitting this theoretical absorption cross-section to the experimental cross-section of the nanoparticle dispersion obtained by spectrophotometry; the result of this fit is shown in Fig. 3, the only adjustable parameter being γ_s. We find $\hbar\gamma_s = 0.012$ eV which corresponds [54] to the damping expected for a nanoparticle of radius 6 nm, in very good agreement with the size of the studied particles. This confinement-corrected gold permittivity (4) with $\gamma = \gamma_0 + \gamma_s$ is used for all the calculations presented in the following.

2.2. Thin hybrid films preparation

A centrifugation step first allows increasing the concentration of the particle aqueous suspension by a factor between 200 and 400, leading to an increase of the optical absorption, monitored by spectrophotometry, with no change of the absorption band shape, which means that, at this step, the nanospheres remain isolated in the suspension. The concentrated nanoparticle suspension is then mixed with a poly(vinyl alcohol) (PVA) (from Mowiol©, of molar mass 195,000 g/mol and hydrolysis ratio 98.4%) solution at a concentration between 1 and 5 wt% and the mixture is deposited on 10×10 mm ^{2} silicon wafers or glass substrates by spin-coating. The volumes of the two components of the mixture and their concentrations are chosen to obtain a given film thickness D, after drying, as well as a given final gold volume fraction f in the hybrid film. Films are finally left under vacuum at room temperature for 24 h to allow complete drying.

In the concentrated mixed suspension before spin-coating, the nanoparticles retain their colloidal stability and do not aggregate, as can be checked by spectrophotometry. The citrate layer surrounding the nanospheres is negatively charged and prevents the clustering of NPs in solution and the merging in case of contact. Clustering – without merging – of the individual nanospheres can occur upon various conditions, like when the electrostatic repulsions between similar nanoparticles are screened in a salt solution [55]. Our studied films all present a homogeneous macroscopic aspect but could present both an inhomogeneous distribution of the nanoparticles and some clustering, possibly occurring during spin-coating.

We have checked, however, that the NPs do keep their spherical shape and do not merge, using TEM imaging of gold nanoparticles scratched from the nanocomposite films (see Fig. 4). These images

show no modification of the particle shapes in the films, when compared to the ones from the dried drop of the synthesis suspension (Fig. 1a).

Atomic force microscopy (AFM) performed in tapping mode with a silicon cantilever of resonance frequency 300 kHz and of tip nominal curvature radius 5 nm, provides top-view topographic images of the films with a resolution of a few nanometers. The images obtained on the studied nanocomposite films are smooth and featureless, which shows that the nanoparticles do not protrude above the film surface. The root mean square roughness over $5 \times 5\ \mu m^2$ is found to increase from 0.1 to 0.3 nm when the gold volume fraction in the films increases from 0 to 1%. These roughness values are, however, always low enough to have no effect on the optical measurements performed on the films.

The presence of the nanoparticles in the film has been confirmed by performing a phase mode image with the amplitude of the AFM tip oscillation optimized so as to probe the difference of hardness beneath the film upper surface [56] (cf. Fig. 5).

3. Optical results

Six samples (labeled S1–S6) are presented in this article, with thicknesses ranging approximately from 30 to 150 nm and gold fractions from 0.8 to 5.7%. Fig. 1b shows a sketch of the samples.

The optical properties of the gold NP-doped polymer films were studied using spectroscopic ellipsometry.

3.1. Spectroscopic ellipsometry

Variable angle spectroscopic ellipsometry was performed in reflection with a UVISEL phase-modulated ellipsometer (from Horiba Scientific) over the spectral range [260–2100 nm] on silicon and [260–800 nm] on glass substrates. We measured $I_s = \sin 2\Psi \sin\Delta$ and $I_c = \sin 2\Psi \cos\Delta$, with Ψ and Δ the two ellipsometric angles, in the UVISEL II (P = 45°;M = 0) configuration, and these data were analyzed using the DeltaPsi2 software from Horiba Scientific. Before the measurements, we checked that no significant depolarization occurred upon reflection on the samples. Depending on the sample, one to five values of the incident angle between 50° and 70° were used, and analyzed simultaneously, and the spot size was either 100 μm or 1000 μm.

Silicon wafers were purchased from Siltronix, France. Superfrost glass slides of thickness 1 mm were used for the samples on glass. Bare silicon and glass substrate as well as five pure poly(vinylalcohol) (PVA) films of different thicknesses ranging from 10 to 300 nm on silicon wafers were measured in the same conditions as the samples. Ellipsometric data measured on the bare silicon substrate were analyzed using the Si and SiO_2 tabulated dielectric functions and yielded a value (1.8 nm) for the native silica layer on the surface. Using these data, the ellipsometric data measured on the five PVA films were fitted to a Fohouri–Bloomer dielectric function (called New Amorphous in the DeltaPsi2 software), providing a full description of the optical index of the polymer matrix used in the calculation of the MG-EMA data. These data are shown on Fig. 6. The ellipsometric data measured on the bare glass substrate taken as a semi-infinite medium were also fitted to a Fohouri–Bloomer dielectric function and used for the analysis of samples S2, S4, S5, and S6.

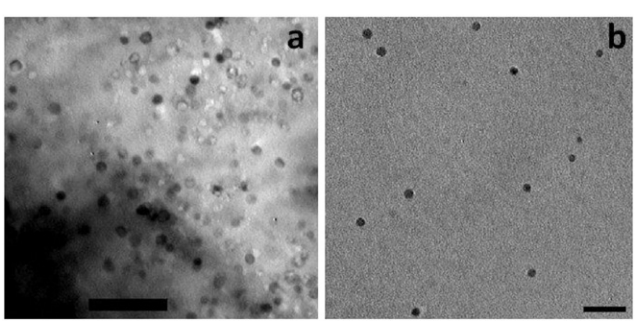

Fig. 4. TEM micrographs of the gold NPs in the films. (a) Image taken on a chip of a nanocomposite film, scratched from the silicon substrate and deposited onto a TEM grid. The dark dots are the nanoparticles, the white bubble-like structures are due to electron beam-induced damage in the polymer. Bar = 200 nm. (b) image taken next to a chip of a nanocomposite film, scratched from the silicon substrate and deposited onto a TEM grid. The dark dots are the nanoparticles, which were detached from the film. Bar = 100 nm.

To extract the complex index of refraction of the films, $\tilde{n} = n + ik$, we classically define a multilayer ellipsometric model (Fig. 7) representing the sample under study with its substrate, whereby indices and thicknesses of all layers are known except those of the nanocomposite film.

In the analysis of spectroscopic ellipsometry data, the unknown optical indices of a sample can be classically obtained either by fitting them with a parametric function or using a direct λ-by-λ inversion appropriate even in the absence of an obvious dielectric function for modeling the sample.

3.2. Maxwell Garnett EMA

The MG-EMA formula is given by [29]:

$$\varepsilon_{\text{eff}} = \varepsilon_m + 3f_{\text{MG}} \cdot \varepsilon_m \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m - f_{\text{MG}}(\varepsilon - \varepsilon_m)}, \tag{7}$$

where ε_{eff} is the effective permittivity of the gold NP-polymer composite, ε_m is the polymer matrix permittivity, ε is the NP gold permittivity, and f_{MG} is the gold volume fraction in the film. The determination of ε used in this formula was discussed in Section 2.1. The polymer matrix permittivity ε_m was determined by ellipsometric measurements performed onto five pure PVA films of different thicknesses ranging from 10 to 300 nm on silicon wafers, prepared in the same conditions as the composite samples. All the pure polymer films could be described by a single Fohouri-Bloomer dielectric function, which is used in Eq. (7), considering that the presence of Au NPs does not modify the dielectric function of the matrix. Moreover, the optical absorption of the PVA films was found equal to zero (see Fig. 6) for the studied wavelength range, which indicates that the absorption observed in the composite films originates only from the Au NPs.

Eq. (7) was applied to the set of measured I_s and I_c values over all measured angles of incidence for each sample. We were unable to obtain good-quality fits, particularly in the region of the localized surface plasmon resonance of the gold nanoparticles. However, a good fit can be obtained in the ultraviolet domain (300–450 nm), as can be seen on Fig. 8.

Two fitting parameters are used: the gold volume fraction (involved in Eq. (7)), and the thickness of the film (involved in the ellipsometric model). The values extracted from the MG-EMA fits in the ultraviolet domain are given in Table 1.

In order to test the validity of these values, we have determined the thicknesses by other means. First, for samples sitting on silicon wafers (S1 and S3), the experimental I_s and I_c data were also adjusted, in the infrared domain (1000–1600 nm), with a Cauchy law $n(\lambda) = A + B/\lambda^2 + C/\lambda^4$ [57] allowing another determination of a thickness D_{Cauchy}. This fit is also presented on the Fig. 8. Thicknesses obtained in both Cauchy and MG approaches always agree closely, as can be seen in Table 1. For samples sitting on glass (S2, S4, S5, and S6), lack of signal in the infrared prevented to apply this test. However, in the case of S4, the MG thickness was compared with the physical thickness obtained from X-ray reflectivity (XRR) measurements and also found in good agreement ($D_{\text{XRR}} = 55.3$ nm vs. $D_{\text{MG}} = 58.6$ nm). Consequently, the thicknesses obtained by the MG fits are reliable, and they will be useful in the inversion procedure.

The MG-EMA fit in the ultraviolet domain also provides values for the gold volume fraction f, which may slightly differ from nominal values, given by the composition of the initial mixed suspension. We

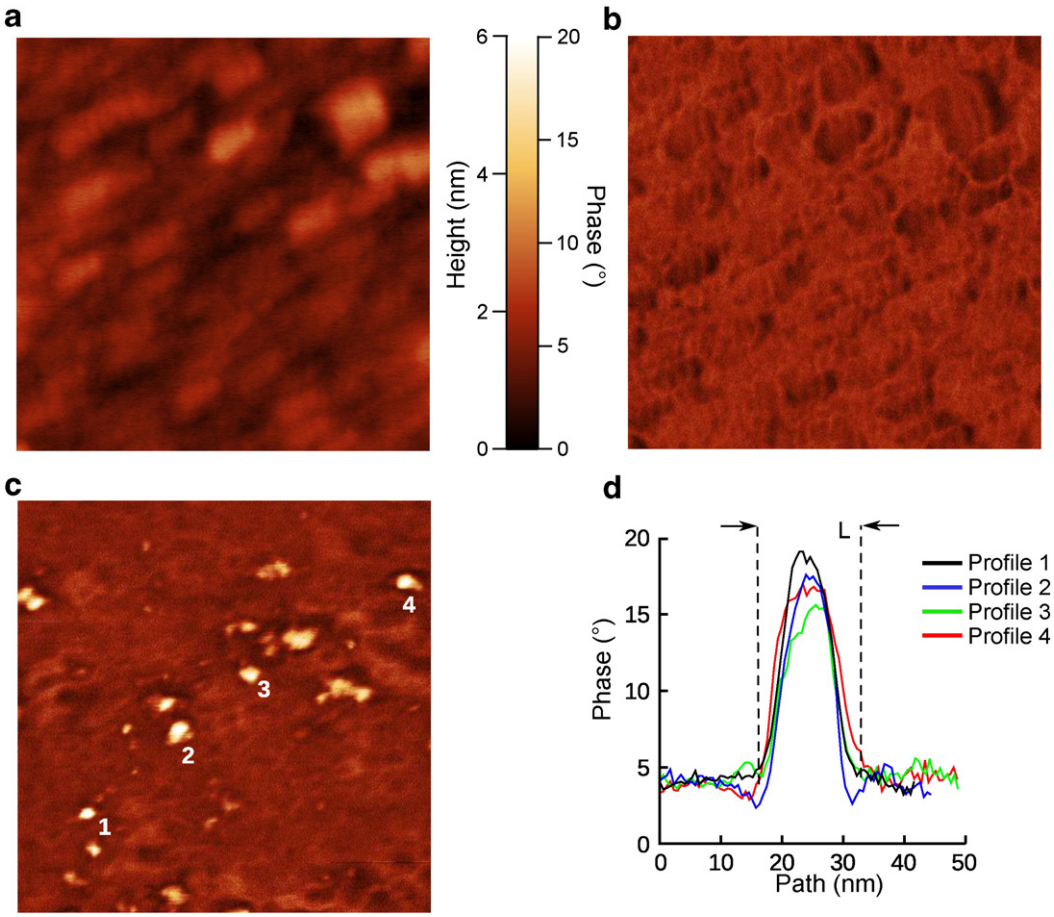

Fig. 5. Atomic force microscopy (AFM) images of a nanocomposite thin film sample. (a) and (b), 600×600 nm ^{2} topography images. (c) 300×300 nm ^{2} phase image with the AFM tip oscillation optimized in order to probe the hardness beneath the upper film surface [56]. The bright spots are nanoparticles beneath the upper film surface. (d) Width of the nanoparticle profile (about 14 nm), consistent with their dimension.

shall emphasize that being able to obtain a reasonably accurate estimate of the gold volume fraction in the sample through fitting the UV ellipsometry data with the MG-EMA, as just explained, is in itself an interesting result: direct experimental measurement of the gold content in the films is uneasy and rather unreliable, because most experimental techniques like plasma ablation, X-ray or electron emission and fluorescence, either require some sort of calibration which can prove difficult in our systems or will provide a value with a large uncertainty.

Finally, accounting for the optical response of the nanocomposite films close to the plasmon resonance wavelength cannot be obtained using the simple Maxwell Garnett EMA.

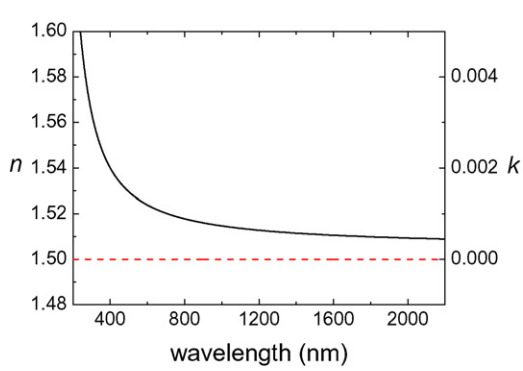

Fig. 6. Refractive index n (continuous line, left axis) and absorption coefficient k (dashed line, right axis, k = 0) for the pure polyvinylalcohol, as determined by a fit of five PVA samples.

3.3. λ-by-λ inversion

The unknown optical indices of the nanocomposite film were then determined using a λ-by-λ inversion, whereby for each wavelength λ in the explored spectral range, n and k are numerically adjusted so as to obtain the best fit to the set of measured I_s and I_c values over all measured angles of incidence. The thickness of the nanocomposite film was taken as the value extracted from the previous MG fit in the UV domain and was used as a fixed parameter.

The real and imaginary parts of the index used for fitting the ellipsometric data in the considered range are given by $\tilde{n} = n + ik = \varepsilon_{\mathrm{eff}}^{1/2}$.

Values of the optical indices n and k of the nanocomposite films, as determined using the λ-by-λ inversion procedure, are displayed as a function of the wavelength in Fig. 9. Samples are presented in order of increasing nanoparticle volume fraction. All six sets of results display a marked resonance close to 540 nm, better visible in the extinction coefficient k spectra. The shape of the resonance can be recognized with a classical S-shape for the n data and a peak for the k data. For the more concentrated samples, the values of n display a significant shift from 1.5 to 1.8 at the resonance. Moreover, we note that most samples present a significant resonance broadening in the red domain. Values for samples S2, S5, and S6 present an additional bump positioned between 650 and 700 nm.

4. Discussion

As was presented before, the comparison between the extracted λ-by-λ data obtained from experimental measurements and the

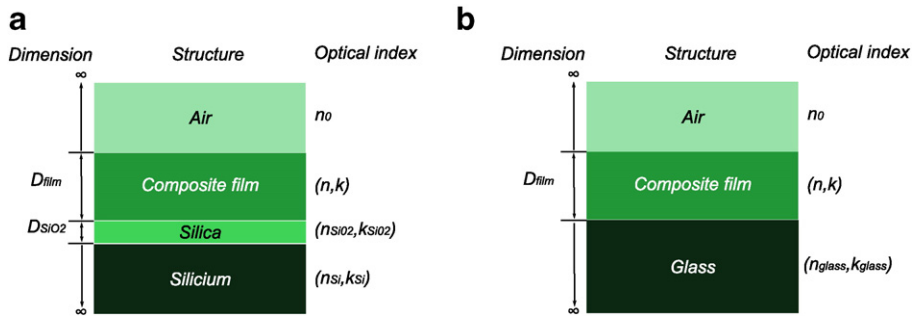

Fig. 7. Representation of the multilayer ellipsometric models. We show both experimental situations with films deposited on (a) a Si/SiO $_2$ substrate and (b) a glass substrate.

MG-EMA results for n and k (Fig. 9) shows a discrepancy in the position, the width and the amplitude of the resonance.

The mismatch lies in the long tails of the experimental resonances and, for several samples, in a secondary peak, seen in the k curves at long wavelength. This may look like a second resonance as can occur for elongated NPs for instance (longitudinal and transverse modes). As we discussed above, however, the shape of the present NPs are clearly spherical and therefore, the occurrence of this shoulder is most likely to be assigned to electromagnetic coupling between nanoparticles. In our samples, where no specific caution has been taken to control the inter-particle distance, it clearly suggests that the distribution of nanoparticles is locally inhomogeneous at the microscopic scale, with some nanoparticles grouping close enough to each other to produce electromagnetic couplings, despite the low average volume fraction.

4.1. Taking interparticle coupling into account: an "equivalent ellipsoid" approach

To improve the fitting of our data and take interparticle coupling into account, we could resort to the large body of literature devoted to extending the applicability of the MG-EMA to more complex samples, based e.g., on spectral density function approaches [35,36,37,38] or on more direct description of particle interactions [39,40,41,42,43,36]. However, these generalized effective medium theories are complex and their application to spectroscopic ellipsometry quickly becomes unpractical.

Another possible approach to describe and fit our experimental results could be to call upon numerical electrodynamical simulations, which either solve directly the Maxwell equations (finite difference time domain, finite elements) or use higher-level approaches like T-matrix [58,59]. Although very powerful, these methods appear unsuitable for our purpose: they all require that the exact spatial distribution of the nanoparticle clusters be given as an input prior to computation, while such a detailed distribution in practice is unknown for any real sample. Moreover, when their number of internal particles becomes in excess of a few units, clusters provide rather unspecific responses, all with similar wide resonances and red-shifted peaks. Therefore, in a complex, many-cluster system, it appears both unrealistic and physically irrelevant to assume one particular distribution of clusters rather than another. Only general features of these distributions are relevant.

Therefore, on the contrary, we propose to take into account coupling effects for spatially close particles in a phenomenological and simplified approach. The result of interparticle couplings is to distort local fields. Rather than considering how complex local field distortions will affect the response of the interacting nanospheres, an *ad hoc* way to consider this situation is to amend the MG-EMA by incorporating the distortions within the nanoparticle polarizabilities themselves. We consider the simplest distortions from spherical polarizabilities, which are ellipsoidal polarizabilities. In other words, instead of considering the complex situation of randomly dispersed, interacting (spherical) nanoparticles with spherical polarizabilities, we simplify the problem by considering the response of an electromagnetically equivalent set of non-interacting particles with ellipsoidal polarizabilities. We call these particles "equivalent ellipsoids," Given the randomness of the nanoparticle positions in the samples, all sorts of interparticle couplings occur; therefore, the set of equivalent ellipsoids needs to include some randomness and a statistical distribution must be called upon.

This phenomenological approach has, however, a physical meaning related to the electromagnetic interactions between close nanospheres. A full theoretical justification is clearly beyond the scope of our present work, but we will show that the ellipsoid polarizabilities are a good model for the optical properties of two close-lying nanoparticles by computing their extinction cross-section using a generalized Mie code. Assuming first that the interaction occurs only through their dipolar fields, it is straightforward to show analytically [60] that the isotropic

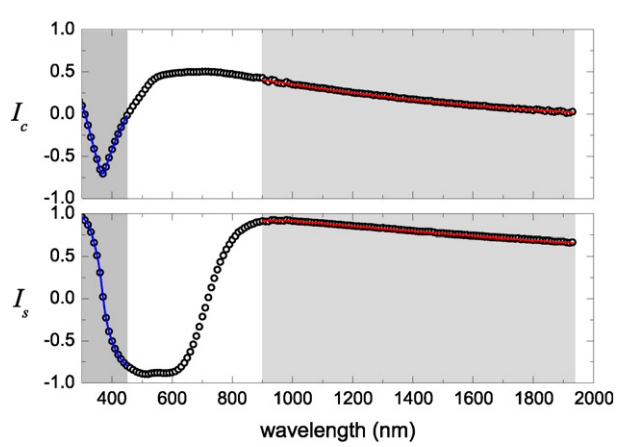

Fig. 8. Plots of the experimental ellipsometric quantities, I_s and I_c, for sample S1. The red curve is the Cauchy fit to the infrared part. The blue curve is the result of the MG-EMA fit in the UV part of the spectrum. The two shaded regions are the restricted spectral domains for the Cauchy and the MG-EMA fits. The non-shaded region is where no obvious fitting function is available.

Table 1
Sample structural parameters. The second column gives the film thickness obtained by fitting the Maxwell Garnett model in the ultra-violet (UV) part of the spectrum. D_{Cauchy} was obtained by fitting a Cauchy dispersion relation in the infrared part of the spectrum. See text for details.

Sample	D_{MG} (nm)	D_{Cauchy} (nm)	f_{MG} (%)
S1	144	141	0.8
S2	75		1.6
S3	82	78	2.4
S4	59		5.5
S5	49		5.4
S6	38		5.7

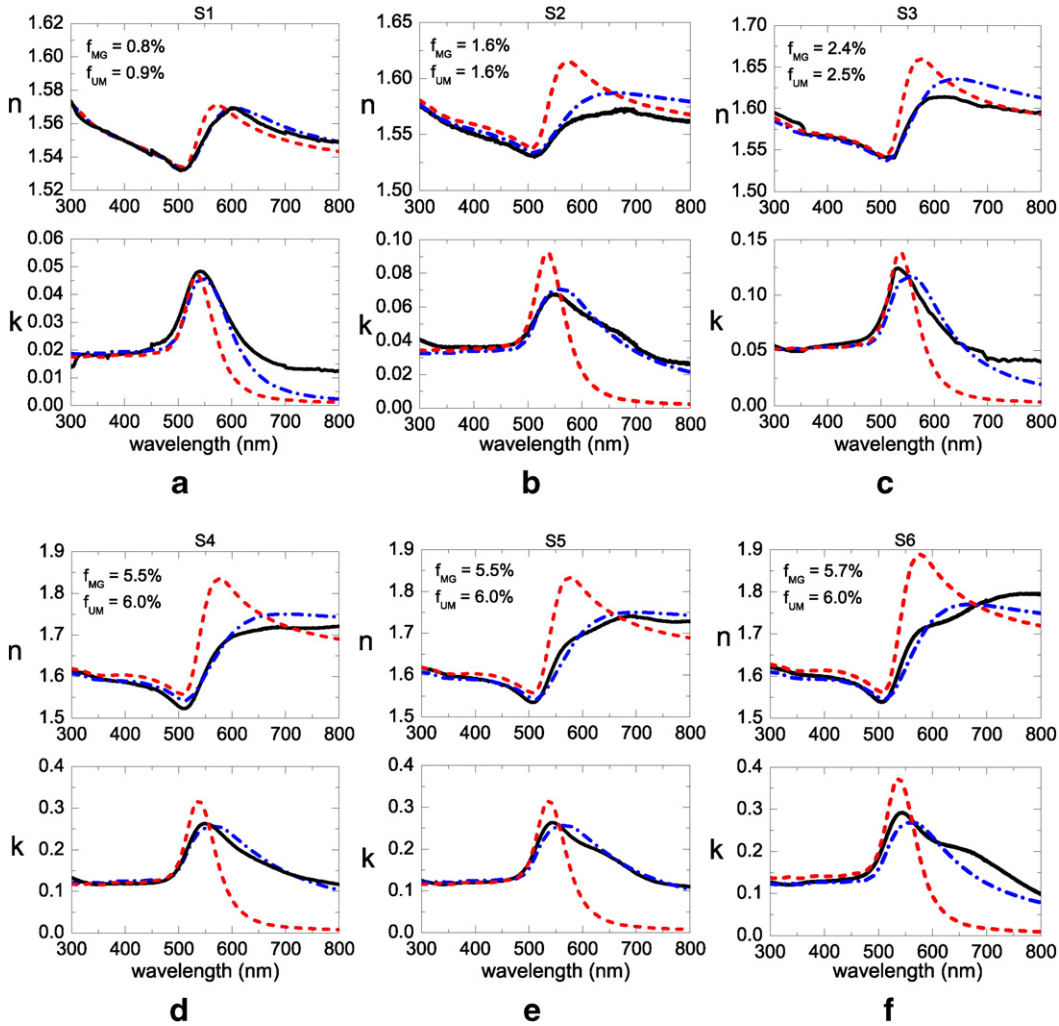

Fig. 9. Optical indices of the studied films. Black continuous curves represent the extracted λ-by-λ data obtained from the experimental measurements, red dashed curves are obtained by the classical Maxwell Garnett formula, and blue dashed-dotted ones are the result of the unimodal equivalent ellipsoid model.

polarizability of each sphere should be replaced by a tensorial one in order to take the coupling into account, given by

$$\alpha^{\mu}_{\text{coupl}} = \frac{\alpha}{1-\alpha\dfrac{F_{\mu}}{4\pi r^3}}, \tag{8}$$

where α is the single sphere polarizability and $\mu = //, \perp$ means that the linear polarization of the incident field is parallel or perpendicular to the axis connecting the two particles. The factors F have the values $F_{//} = 2$ and $F_{\perp} = -1$, because in the parallel case, the dipolar electric field radiated by one sphere and acting on the other one is twice and in opposite direction with respect to the dipolar moment in the perpendicular case. If we substitute in Eq. (8), the classical expression for the single sphere polarizability

$$\alpha = 4\pi R^3 \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m} \tag{9}$$

we obtain after some rearrangements:

$$\alpha^{\mu}_{\text{eff}} = 4\pi R^3 \frac{\varepsilon - \varepsilon_m}{3\varepsilon_m + 3L_{\mu}(\varepsilon - \varepsilon_m)}, \tag{10}$$

where

$$L_{\mu} = \frac{1}{3}\left[1 - F_{\mu}\left(\frac{R}{r}\right)^3\right], \tag{11}$$

r being the distance between the spheres centers. Furthermore, we note that

$$L_{//} + 2L_{\perp} = 1 \tag{12}$$

We therefore see that Eq. (10) is indeed formally identical to the polarizability of an ellipsoid [48] with depolarization factors L_{μ} and volume equal to the single sphere volume. This directly supports the idea that a pair of coupled particles may be seen as two uncoupled particles with equivalent ellipsoidal polarizabilities. Note that, within this dipolar approximation, the equivalent ellipsoid is always spheroidal.

This formal identification being only true in the dipolar regime of coupling, in a second step, we incorporate multipoles as they enter into action when interacting particles become close enough. We did this numerically using Mackowski's T-matrix code [58,59], which allowed us to compute the scattering properties of an interacting pair based on a generalized Mie code. The extinction cross-section of interacting spheres within a pair (a "bisphere"), where up to 30 multipoles were taken into account for each particle, was calculated for both polarizations of the incoming field (parallel to the pair center-to-center axis, or perpendicular). We chose a bisphere formed by two 6.5 nm radius spheres separated by 0.4 nm, i.e. almost touching spheres. The result is drawn in Fig. 10. In this case, two resonances are obtained for parallel polarization, and one for perpendicular polarization.

We observe that the perpendicular response of the bisphere is close to that of the single sphere, while the parallel response is the sum of an intense peak at longer wavelength and of a smaller one, slightly red-shifted with respect to the one of a single sphere. This shows that a bisphere crudely consists of the optical response of two single spheres, with an additional longitudinal resonance at longer wavelength. Such behavior (short and long wavelength resonances) appears to be similar to what is expected for an elongated ellipsoid. In case of systems with disoriented bi-spheres, the extinction cross-section would be an average of the parallel and perpendicular cross-sections (not shown here). Consequently, this exact calculation is now to be compared to the (analytical) extinction cross-section obtained for a collection of disoriented ellipsoids which is directly proportional to the polarizabilities α_i along the principal axes [48]:

$$\sigma_{\text{ext}} = \frac{2\pi}{3\lambda}\text{Im}(\alpha_1 + \alpha_2 + \alpha_3), \tag{13}$$

where an isotropic average has been taken out by adding all contributions together, so as to reflect the random orientation of the disoriented ellipsoids, and where λ is the wavelength of light in the medium surrounding the particles.

The resulting resonance curve for the randomized ellipsoid (with depolarization factors chosen equal to $L_1 = 0.15$, $L_2 = 0.3$ and $L_3 = 1 - L_1 - L_2$) is traced in Fig. 10. With such parameters, the main peak of the parallel response is correctly reproduced by the ellipsoidal cross-section curve. On the other hand, the small wavelength resonances for the bisphere are also correctly reproduced by the ellipsoidal cross-section, but with smaller intensity. We conclude that, at least for the simple case of two-body interaction, the actual optical response, and primarily its main red-shifted peak, can be reasonably well represented by an ellipsoidal analogue, or "equivalent ellipsoid." In order to reproduce the intensity of the short-wavelength resonance, close to the one of the single sphere, an additional component corresponding to a sphere would be needed.

This study leads us to extend the "equivalent ellipsoid" approach to now describe real systems. Real samples contain all types and shapes of clusters beyond simple pairs as considered before, and include also single spheres. In the following of this article, we take on a phenomenological point of view by using the "equivalent ellipsoid" approach with our samples and discussing how well it applies.

4.2. Modified Maxwell Garnett EMA with a unimodal distribution of equivalent ellipsoids

To extend the classical MG-EMA formula, we turn to a general form able to include ellipsoidal polarizabilities with some polydispersity

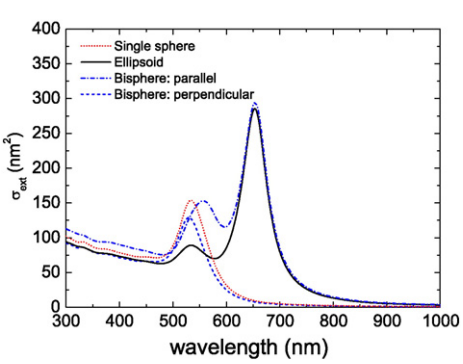

Fig. 10. Comparison between a bisphere and an ellipsoid. Extinction cross-sections for a single sphere (red dotted curve), a randomized ellipsoid with $(L_1,L_2) = (0.15, 0.3)$ (black), and a bisphere (dash-dotted blue curve, parallel polarization; dashed blue curve, perpendicular polarization) with a 13.4 nm center-to-center separation and 6.5 nm sphere radii.

(distribution of size, shape, orientation). In situations with real ellipsoidal inclusions, this modification to MG has long been proposed by Galaneer, by Cohen et al., then by Bohren and Huffman [61,62,48], and has been studied further on a theoretical level by Goncharenko et al. [63,64,65]. In contrast with these previously mentioned works, however, we use this idea here, in an "equivalent" approach, to describe couplings between randomly distributed spheres: the ellipsoids, which will be introduced now, are not material but they represent inclusions with ellipsoidal polarizabilities, which are electromagnetically approximately equivalent to the set of interacting spherical particles in our samples.

One generalized MG formula for a distribution of ellipsoidal inclusions is given by [48]

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_m(1-f) + \beta f \varepsilon}{(1-f) + f\beta}, \tag{14}$$

where ε_m and ε are the dielectric permittivities of the matrix and the particles respectively and f the volume fraction of the particles in the sample. The factor β introduces the statistical distribution of equivalent ellipsoids. By assuming that the shape, volume, and orientation of the equivalent ellipsoids are statistically uncorrelated properties, we obtain the following expression:

$$\beta = \int\int P(L_1, L_2)\frac{1}{3}\sum_{j=1}^{3}\frac{\varepsilon_m}{\varepsilon_m + L_j(\varepsilon - \varepsilon_m)}\text{d}L_1\text{d}L_2, \tag{15}$$

where L_1 and L_2 are the two ellipsoidal depolarization factors [66], with $L_1 + L_2 + L_3 = 1$, and $P(L_1,L_2)$ is the shape probability distribution of the equivalent ellipsoids, which is a function of the depolarization factors L_1 and L_2. Note that for spheres, we have $L_1 = L_2 = L_3 = 1/3$, $P(L_1,L_2)$ is a delta function at $(1/3, 1/3)$ and the factor β reduces to

$$\beta = \frac{3\varepsilon_m}{\varepsilon + 2\varepsilon_m}. \tag{16}$$

which, together with Eq. (14), leads to recovering the usual MG-EMA formula of Eq. (7) for spheres dispersed in a dielectric matrix.

The next step is to determine the equivalent ellipsoid shape distribution $P(L_1,L_2)$. In this section, we consider only a unimodal distribution, i.e. a distribution centered around one shape of particles. Because we consider that the ellipsoidal polarizabilities are due to electromagnetic couplings of a population of material spheres, we start with the assumption that physically, the shape distribution of the equivalent ellipsoids should be centered around spheres (i.e., assuming that spherical response will be most probable):

$$P(L_1, L_2) = P_s(L_1, L_2) \tag{17}$$

where $P_s(L_1,L_2)$ has a maximum for $(L_1,L_2) = (1/3, 1/3)$ as emphasized by the subscript "s." Goncharenko has studied several variants [64,63] of such distributions. We here decided to use the Gaussian-type distribution:

$$\begin{aligned} P_s(L_1, L_2; c) &= N(c)\exp[-cL_1(L_1-1)\exp -cL_2(L_2-1)] \\ &\times \exp[-c(L_1+L_2)(L_1+L_2-1)], \end{aligned} \tag{18}$$

where $N(c)$ is a normalization constant and c is the parameter which determines the width of the distribution. Note that the higher the value of c, the narrower the distribution. In Fig. 11, we represented the domain in which $P_s(L_1,L_2)$ is defined together with a plot of the function.

As can be seen from Fig. 11a, P_s includes contributions of ellipsoids with all shapes in the (L_1,L_2) plane, where the specific lines in the plane correspond to the prolate and oblate symmetries. Fig. 11b gives one example of the shape of P_s. Finally, Fig. 11c proposes an alternative plot of P_s applicable to the prolate and oblate situations, for which it is possible to express the probability function P_s directly in terms of the

(more intuitive) ellipsoid aspect ratio a/b (with a and b being the small and large axis of the equivalent ellipsoid respectively), rather than the more abstract depolarization factors L_1, L_2.

In Fig. 9, we superimposed with the experimental data both the results of the usual MG-EMA (Eq. (7)) and the ones obtained with the equivalent ellipsoid MG-EMA (Eq. (14)) just described.

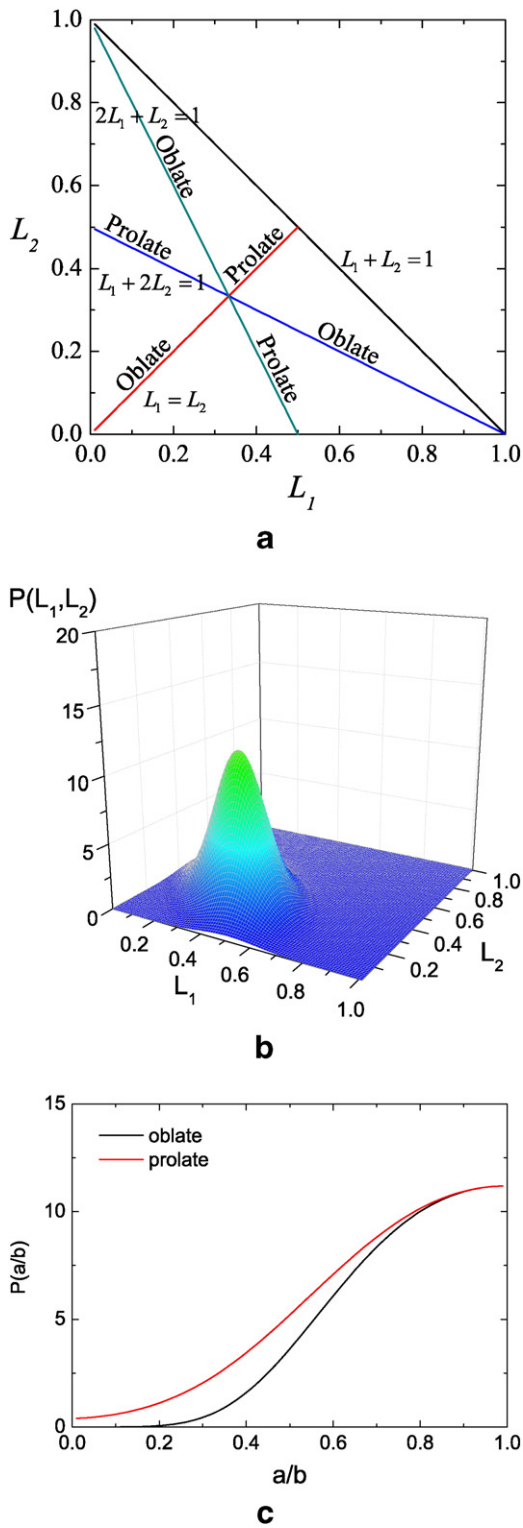

Fig. 11. Ellipsoid shape distribution. (a) Domain of definition of the function $P_s(L_1,L_2)$ (i.e., under the $L_1+L_2=1$ segment). (b) Plot of $P_s(L_1,L_2)$ for $c=20$. (c) Plot of $P_s(a/b)$ for prolate and oblate spheroids with $c=20$.

The values of the parameters used to fit the data are given in Table 2. Note that for the classical MG-EMA, only one parameter, the volume fraction f_{MG} of gold NPs, has to be fitted. For the equivalent ellipsoid MG-EMA, two parameters are fitted: the volume fraction of gold f_{UM} and the width c of the ellipsoid distribution P, where UM stands for unimodal.

As can be seen from Fig. 9, the equivalent ellipsoid model fits the experimental data much better than the MG-EMA formula, with both plasmon amplitude and the long-wavelength spectrum better reproduced. Satisfactorily, for all studied samples, the MG and equivalent ellipsoid EMAs give very similar gold volume fractions. This can be qualitatively explained by the fact that the integrated intensity of the plasmon resonance is approximately proportional to the amount of gold and depends only weakly on the shape distribution of the particles. Finally, the values of the distribution width parameter c correlate with the gold volume fraction: higher volume fractions correspond to lower values of c, which means wider shape distributions and larger aspect ratio of the equivalent ellipsoids. This is in agreement with the fact that the aspect ratio is a representation of the strength of the interparticle couplings.

Still, it can be observed that the equivalent ellipsoid model as introduced above has some limitations: as the model is made to reproduce the broadening of the plasmon resonance under the effect of interparticle couplings, it also introduces a red-shift of the resonance maximum, more pronounced than is observed in the experimental data. This indicates that our model underestimates the number of purely spherical particles in the samples (i.e., particles located at the maximum of the probability density function corresponding to spheres weakly interacting with others). Furthermore, on samples S2, S4, S5, and S6, the secondary resonance (shoulder) observed at long wavelengths is not reproduced, suggesting the presence of a second population of equivalent ellipsoids corresponding to more strongly coupled particles.

We nevertheless believe that this unimodal model is valuable in itself as it provides a simple and yet efficient description of random systems like ours, with the help of only a minimal number of fitting parameters; depending on applications, this may be good enough. In the next session, we will turn to a bimodal equivalent ellipsoid model, which will prove more precise in terms of fitting, but will naturally lose in simplicity.

4.3. Modified Maxwell Garnett EMA with a bimodal distribution of equivalent ellipsoids

In order to decorrelate, in the resonance, the broadening of the peak from its position, and also to model the secondary resonance, the most natural modification to the previous model is to introduce a second population P_e of equivalent ellipsoids. (Additional distributions could in principle be introduced as well for better fits, only at the expense of multiplying fit parameters; we will show that two distributions are already enough to obtain satisfactory fits.)

The distribution P_e is chosen centered around specific depolarization factors $(L_1,L_2)\neq(1/3,1/3)$. In other words, P_e is centered on a most probable ellipsoid, as denoted by the subscript "e," rather than a sphere as was the case for P_s. This new population P_e should be understood as that of strongly interacting spherical NPs in close vicinity to each other, as we showed above for the case of a pair of spheres.

Table 2
Values of the parameters used to obtain the fits in Fig. 9 (UM stands for unimodal).

Sample	f_{MG} (%)	f_{UM} (%)	c
S1	0.83	0.9	60
S2	1.62	1.62	20
S3	2.42	2.5	30
S4	5.48	6.0	15
S5	5.45	6.0	15
S6	5.71	6.0	20

We therefore now introduce a bimodal probability function for the equivalent ellipsoids, including both types of populations:

$$P(L_1, L_2) = w_s\, P_s(L_1, L_2) + w_e\, P_e(L_1, L_2) \tag{19}$$

The coefficients w_s and w_e are the statistical weights for each distribution with $w_s + w_e = 1$.

The secondary distribution P_e, which accounts for the secondary resonance, has the same functional form as P_s (see Eq. (18)), only translated by the translation factors p_1 and p_2:

$$P_e(L_1, L_2; c_e) = P_s(L_1 - p_1, L_2 - p_2; c_e). \tag{20}$$

Since the spatial orientations of our equivalent ellipsoid were assumed to be random in our model, there are in fact several equivalent choices of values for p_1 and p_2, which will translate the initial distribution into different regions of the (L_1,L_2)-plane (see Fig. 11) with exactly the same optical properties: we took p_1 and p_2 negative, without any loss of generality.

In Fig. 12, a 3D plot of an example of bimodal distribution $P(L_1,L_2)$ is given, as well as sections for prolate and oblate spheroids as a function of the aspect ratio a/b. The bimodal distribution shows here two peaks, one centered on the spheres ($L_1 = L_2 = 1/3$), the other centered on a mean ellipsoid that permits us to describe correctly the observed optical spectra. The first one corresponds to the maximum of $P(a/b)$ for $a/b = 1$, the second one to the maximum around $a/b = 0.46$ for the oblate spheroids.

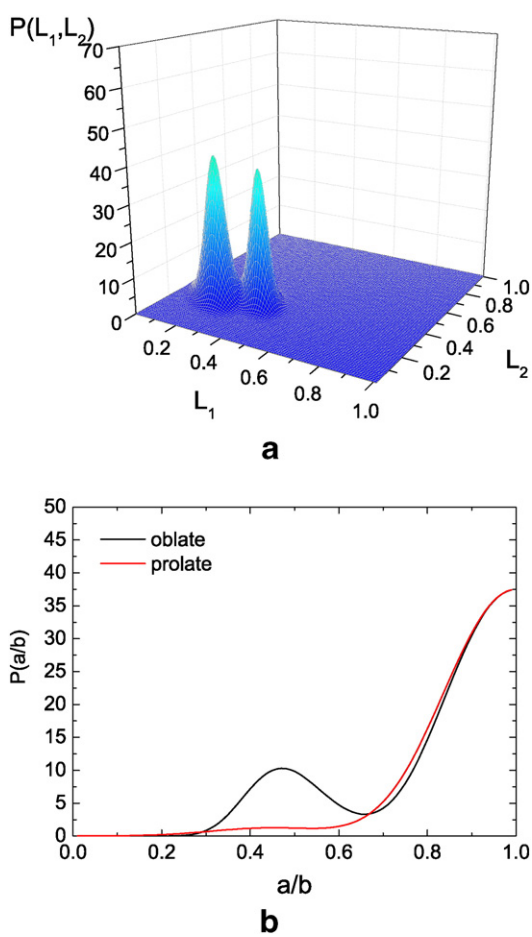

Fig. 12. Distribution used for fitting sample S6. (a) Plot of $P(L_1,L_2)$ for $c_s = 170$, $c_e = 120$, $w_s = 0.4$, and $w_e = 0.6$. b) Plot of $P(a/b)$ for prolate and oblate spheroids.

In this bimodal equivalent ellipsoid model, there are now 6 parameters to fit (in contrast with the unimodal model, which had only 2): the volume fraction f_{BM} (BM stands for bimodal), the width parameters c_s and c_e of P_s and P_e, respectively, the position of the maximum (i.e., most probable ellipsoid) in the (L_1,L_2)-plane for the second distribution $L_{1,e}^{max} = \frac{1}{3} + p_1$ and $L_{2,e}^{max} = \frac{1}{3} + p_2$, and finally, one of the statistical weights (w_s). The bimodal model was applied to our experimental samples: Fig. 13 shows the results of these fits and Table 3 gives the fit parameter values.

Qualitatively, it can firstly be seen that the excessive red-shifting of the main resonance produced in the unimodal model has been corrected, now coinciding much better with experiments. Secondly, it is also now possible to reproduce the secondary resonance and the long tail for samples S2, S4, S5, S6, with very good accuracy for the k curves, and reasonable accuracy for n curves. Note that the fits improve also for the two other samples (S1, S3), where a secondary resonance is not apparent. To allow for a quantitative comparison, mean-square errors (MSE) between fits and data were calculated (see Table 3): for all samples but S2 and S5, bimodal fits are confirmed to be significantly more accurate than unimodal ones. Even if the improvement in S2 and S5 is not numerically significant, due to a discrepancy in the larger wavelength region, the resonance region is still better fitted by the bimodal EMA (especially for k curves).

From Table 3, we also remark that for the samples S2, S5, and S6, which present a clear secondary resonance peak, and for S4 which shows an intense absorption tail for large wavelengths, we have $w_e > w_s$. For the other samples (S1, S3), where this bump is not as visible, or totally absent, we have $w_e < w_s$. The presence of the bump indeed denotes the existence of a significant population of spherical particles in situations of strong coupling, and satisfyingly enough, this occurs primarily for the more concentrated samples.

Finally, Table 3 provides additional data on the second population P_e. For each sample, the position of the maximum of P_e in the (L_1,L_2)-plane is given: $L_1 = L_{1,e}^{max}$ and $L_2 = L_{2,e}^{max}$. Also, the resonance position for each distribution is specified: for P_s, the resonance peak is of course that of the sphere on which it is centered, $\lambda_s^{res} = 532$ nm. For P_e, the most probable ellipsoid with depolarization factors $(L_{1,e}^{max}, L_{2,e}^{max})$ has three resonances; the resonance position λ_e^{res} given in Table 3 corresponds to the most redshifted one, which dominates the others and is indeed the one generating the secondary shoulder peak in the extinction coefficient curves, as can be seen by comparing the values for λ_e^{res} and the curves in Fig. 13.

At this point, it is interesting to backtrack to the T-matrix simulation results displayed in Fig. 10, which can provide some qualitative information on the degree of coupling experienced in particles within the second population P_e: the plotted ellipsoidal resonance curve corresponds to that of the most probable ellipsoid in population P_e in sample S6, with the most probable values $(L_{1,e}^{max}, L_{2,e}^{max}) = (0.15, 0.3)$. In Fig. 10, the resonance peak of this ellipsoid coincides with the largest peak of a bisphere with a particle surface-to-surface separation of 0.4 nm. This means that the particles corresponding to the most probable value in P_e are essentially in contact. This separation is smaller than the one (around 1 nm) observed for an equivalent system [67]. This discrepancy may be related to the fact that in this simple calculation, we only considered bispheres; chains of spheres or larger aggregates of spheres would shift the secondary resonance to longer wavelength, and the agreement with experiment would therefore need a larger separation between particles. In any case, this confirms that the second population P_e indeed corresponds to a population of strongly interacting, near-contact particles, in contrast to the first population P_s which corresponds to rather weakly interacting particles.

We point out that we have, so far, not found an experimental technique capable of evidencing these two electromagnetically different populations in the samples in terms of structural features, but it could be the subject of further studies.

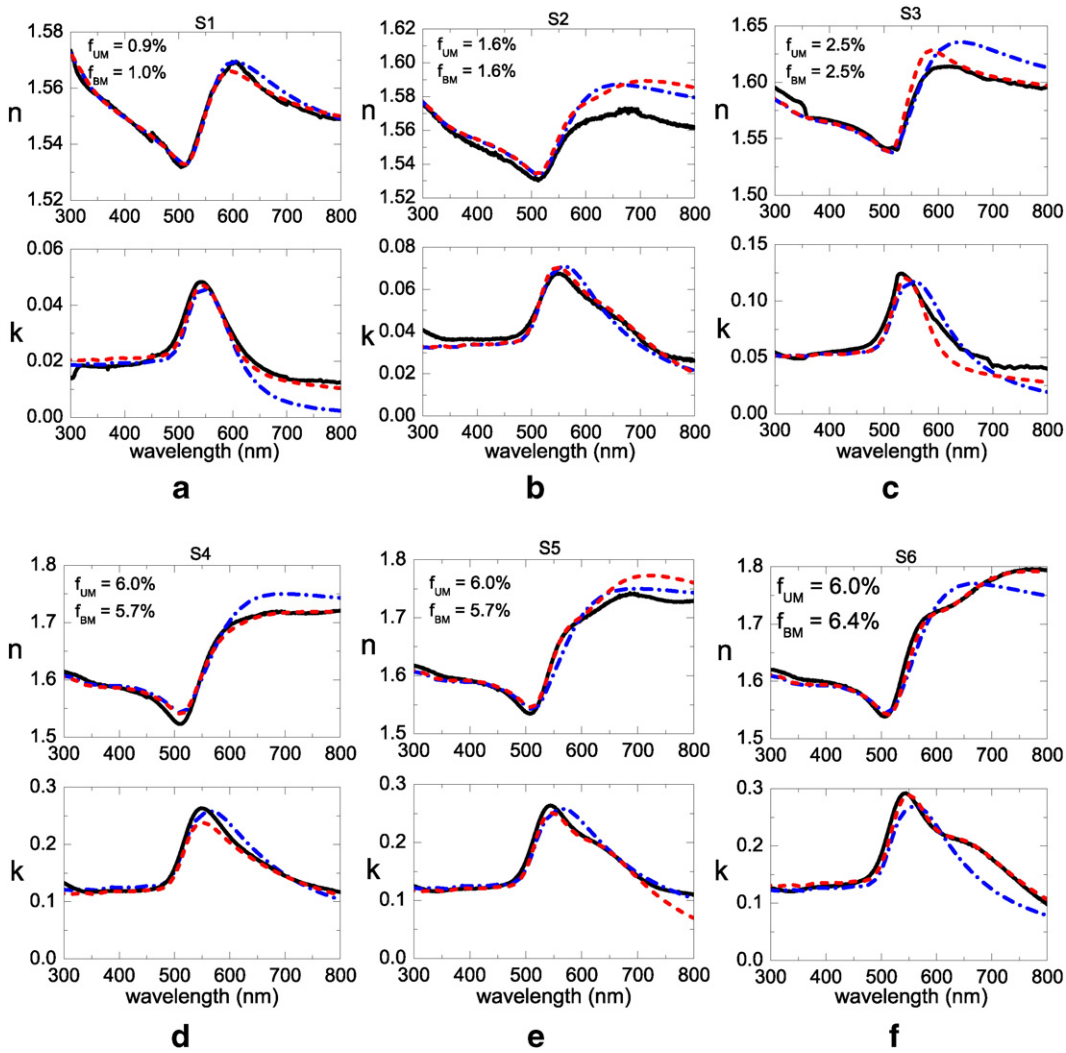

Fig. 13. Optical indices of the studied films. Black continuous curves represent the experimental data, blue dashed-dotted curves are obtained by the unimodal equivalent ellipsoid model, and red dashed ones are the result of the bimodal model.

5. Conclusions

We have fabricated and characterized polymer films doped with gold nanospheres with a volume fraction between 1% and 5%, which we studied as a model system for disordered self-assembled plasmonic nanomaterials. Using spectroscopic ellipsometry, we have extracted the effective optical properties of these films. We showed that even for such low-volume fractions, the effective optical properties are not well reproduced by the classical Maxwell Garnett model. This is due to the presence of concentration heterogeneity in the samples, generating electromagnetic couplings between some of the particles. To model these, we turned here to a generalization of the MG-EMA. The spherical nanoparticles are considered to respond with various "equivalent" ellipsoidal polarizabilities, either with a unimodal or a bimodal shape distribution, to account for the spatial distribution of interparticle electromagnetic couplings within the samples. We showed that this EMA can be quite effective to describe the actual samples with strong interaction between particles. It presents the advantage of being easy to implement, compared to more detailed approaches cited earlier, and much less time-consuming: this can make it a useful material dispersion law for use in experimental curve-fitting procedures in ellipsometry, for example. Another huge advantage is that being phenomenological in nature, our approach does not require any assumption on the detailed nanoparticle organization and/or exact cluster distribution within the samples, which are out of reach experimentally but must nonetheless be assumed to implement full electrodynamic simulations. Such modified Maxwell Garnett EMA could be a general tool for the description of plasmonic couplings in different disordered nanocomposites.

Table 3
Values of the model parameters used to obtain the fits in Figs. 9 and 13: notations are defined in the main text.

Sample	f_{MG} (%)	f_{BM} (%)	w_s	w_e	c_s	c_e	$L_{1,e}^{max}$	$L_{2,e}^{max}$	λ_s^{res} (nm)	λ_e^{res} (nm)	MSE UM	MSE BM
S1	0.83	1.0	0.7	0.3	100	20	0.13	1/3	532	681	1.74×10^{-5}	3.47×10^{-6}
S2	1.62	1.62	0.43	0.57	80	120	0.16	1/3	532	631	6.83×10^{-5}	8.45×10^{-5}
S3	2.42	2.5	0.67	0.33	200	30	0.13	1/3	532	681	1.6×10^{-4}	7.6×10^{-5}
S4	5.48	5.7	0.2	0.8	300	20	0.19	0.31	532	605	2.36×10^{-4}	7.73×10^{-5}
S5	5.45	5.7	0.33	0.67	200	120	0.17	1/3	532	624	1.39×10^{-4}	2.58×10^{-4}
S6	5.71	6.4	0.4	0.6	170	120	0.15	0.3	532	649	5.87×10^{-4}	6.93×10^{-5}

Acknowledgments

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